

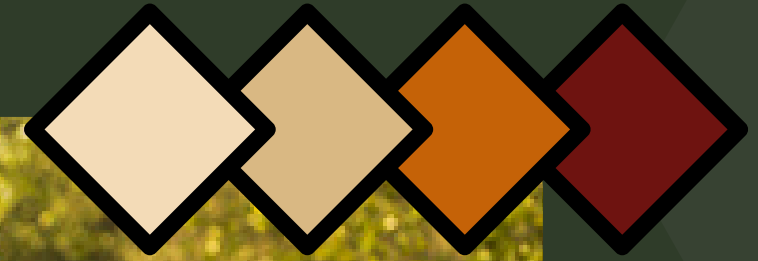
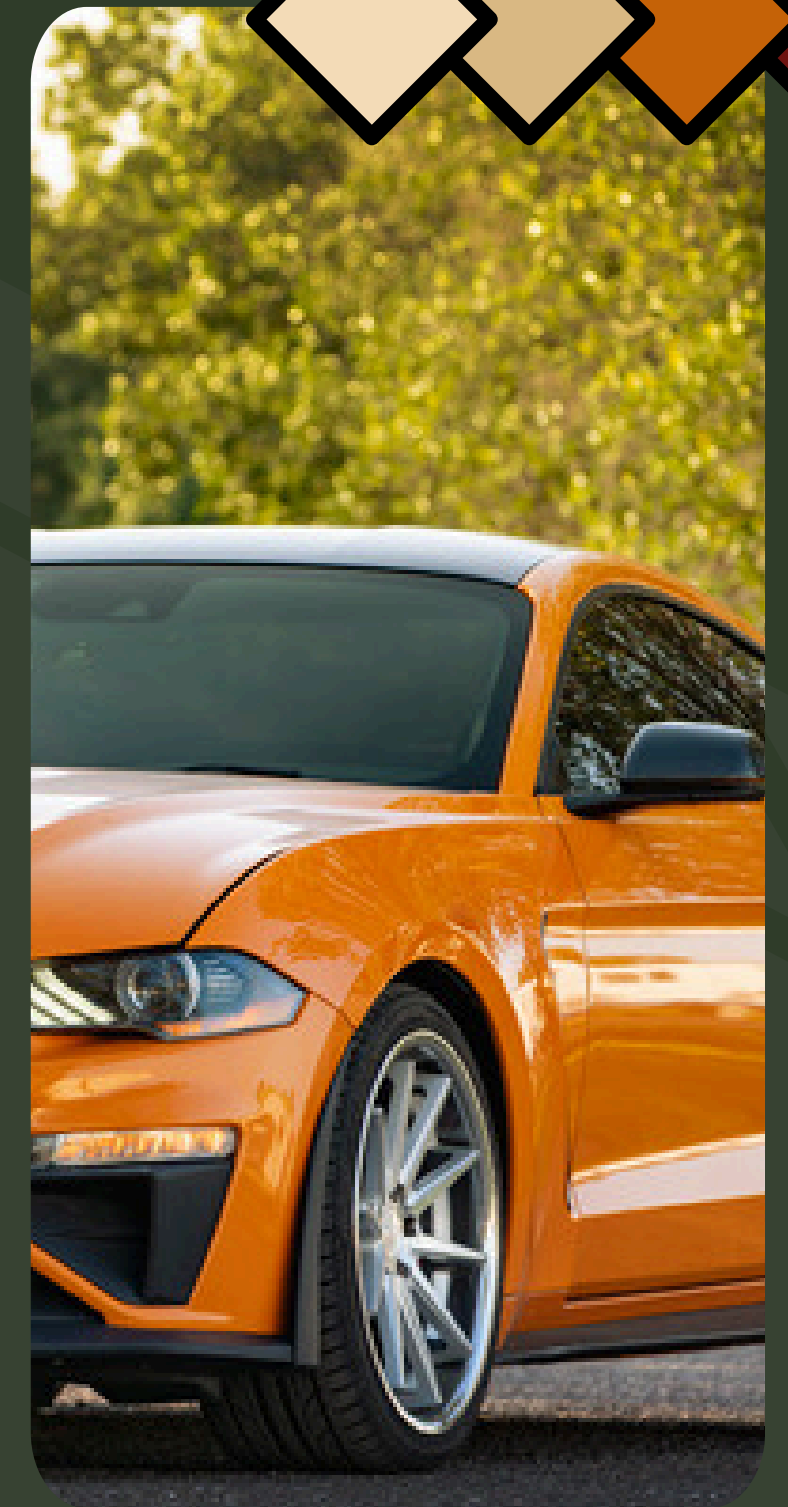
Clustering and Fitting

Dataset : Electric Vehicle Population Data
(Data on the number of Electric Vehicles
on the road of United states)

Dataset Source : [Kaggle](#)

Github : [Link to Repository](#)

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Dataset Overview

The Electric Vehicle Population Dataset consists of 112,634 records and 17 columns, each representing vehicle registration details, manufacturer information and electric specifications for electric vehicles in Washington State, USA.

Column Name	Type	Description
VIN (1-10)	object	Vehicle's unique identification number prefix
County	object	County where vehicle is registered
City	object	City of electric vehicle registration
State	object	U.S. state of vehicle registration
Postal Code	int64	ZIP code of registered vehicle location
Model Year	int64	Year the vehicle was manufactured
Make	object	Electric vehicle manufacturer or brand
Model	object	Specific electric vehicle model name
Electric Vehicle Type	object	Battery or plug-in hybrid classification
Clean Alternative Fuel Vehicle (CAFV) Eligibility	object	Indicates clean-fuel program eligibility status
Electric Range	int64	Maximum miles on full electric charge
Base MSRP	int64	Manufacturer's suggested retail vehicle price
Legislative District	object	Washington state legislative district number
DOL Vehicle ID	int64	Department of Licensing vehicle identifier
Vehicle Location	object	Geographic coordinates of vehicle registration
Electric Utility	object	Electricity provider serving registered address
2020 Census Tract	int64	Census tract code of registration area

Data Preparation

- Imported essential Python **libraries**, mounted Google Drive, and **loaded** the Electric Vehicle Population **Dataset**; previewed structure, data types, and non-null counts for initial understanding.
- **Dropped unnecessary columns** (Postal Code, Legislative District, Electric Utility, 2020 Census Tract) to simplify and streamline the dataset.
- **Renamed selected columns** for better clarity and consistency:
Clean Alternative Fuel Vehicle (CAFV) Eligibility → `cafv_eligibility`,
Make → `manufacturer`.
- Identified both **numerical and categorical columns**, converting categorical ones to the 'category' datatype for optimized performance.
- **Handled missing values** effectively by filling Model with its mode and dropping rows with missing Vehicle Location data.
- **Replaced 0 values with the overall median** for Base MSRP and with type-specific median for Electric Range to ensure accurate numerical data.
- **Extracted accurate Latitude and Longitude** coordinates from Vehicle Location for further spatial analysis.
- Checked thoroughly for **duplicate rows** and confirmed none existed, ensuring clean and reliable data.

Statistical Depth

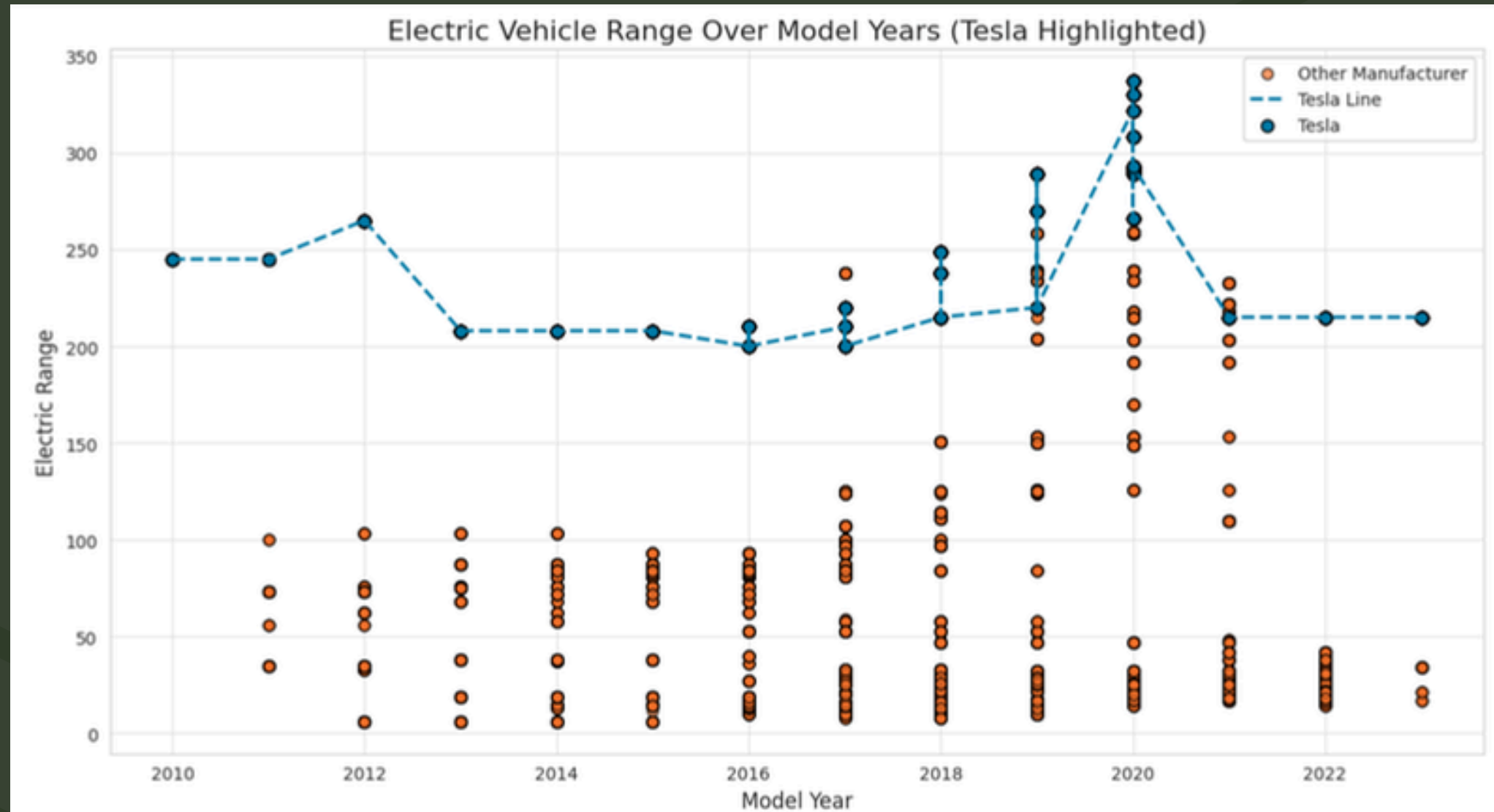
Numeric Insights

- **Model Year:** Average ≈ **2019** (range: 1997–2023) — most vehicles are recent.
- **Electric Range:** Average ≈ **162 miles** (range: 6–337) — wide variation in battery performance.
- **Base MSRP:** Average ≈ **\$59,826** (range: \$31,950–\$845,000) — shows both affordable and luxury vehicles.
- **DOL Vehicle ID:** Unique identifiers, mean ≈ **199 million**.
- **Latitude & Longitude:** Mostly around **lat -121, lon 47**, reflecting concentration in Washington State.

Categorical Insights

- **Manufacturer:** **TESLA** dominates with 52,074 vehicles (≈46% of dataset).
- **Model:** **MODEL 3** is most common (23,153 vehicles).
- **Electric Vehicle Type:** **Mostly Battery** Electric Vehicles (BEV: 86,030); fewer Plug-in Hybrid Electric Vehicles (PHEV: 26,580).
- **CAFV Eligibility:** Majority **eligible** (58,626 of 112,610 vehicles).
- **Geography:** Concentrated in **Seattle & King County**; 758 unique vehicle locations across 45 states.
- **VIN Prefixes:** **7,548** unique prefixes; most common is 5YJYGDEE9M.

Relational Graphs - scatter plots with trend lines



1 ↓

```
# Filter dataset from start_year
df_filtered = df[df['Model Year'] >= start_year].copy()
df_filtered['manufacturer'] = df_filtered['manufacturer'].str.title()
```

2 ↓

```
# Non-Tesla points
df_non_tesla = df_filtered[df_filtered['manufacturer'] != 'Tesla']
plt.scatter(
    df_non_tesla['Model Year'],
    df_non_tesla['Electric Range'],
    color='#f3722c',
    alpha=0.7,
    label='Other Manufacturer',
    s=40,
    edgecolor='black'
)
```

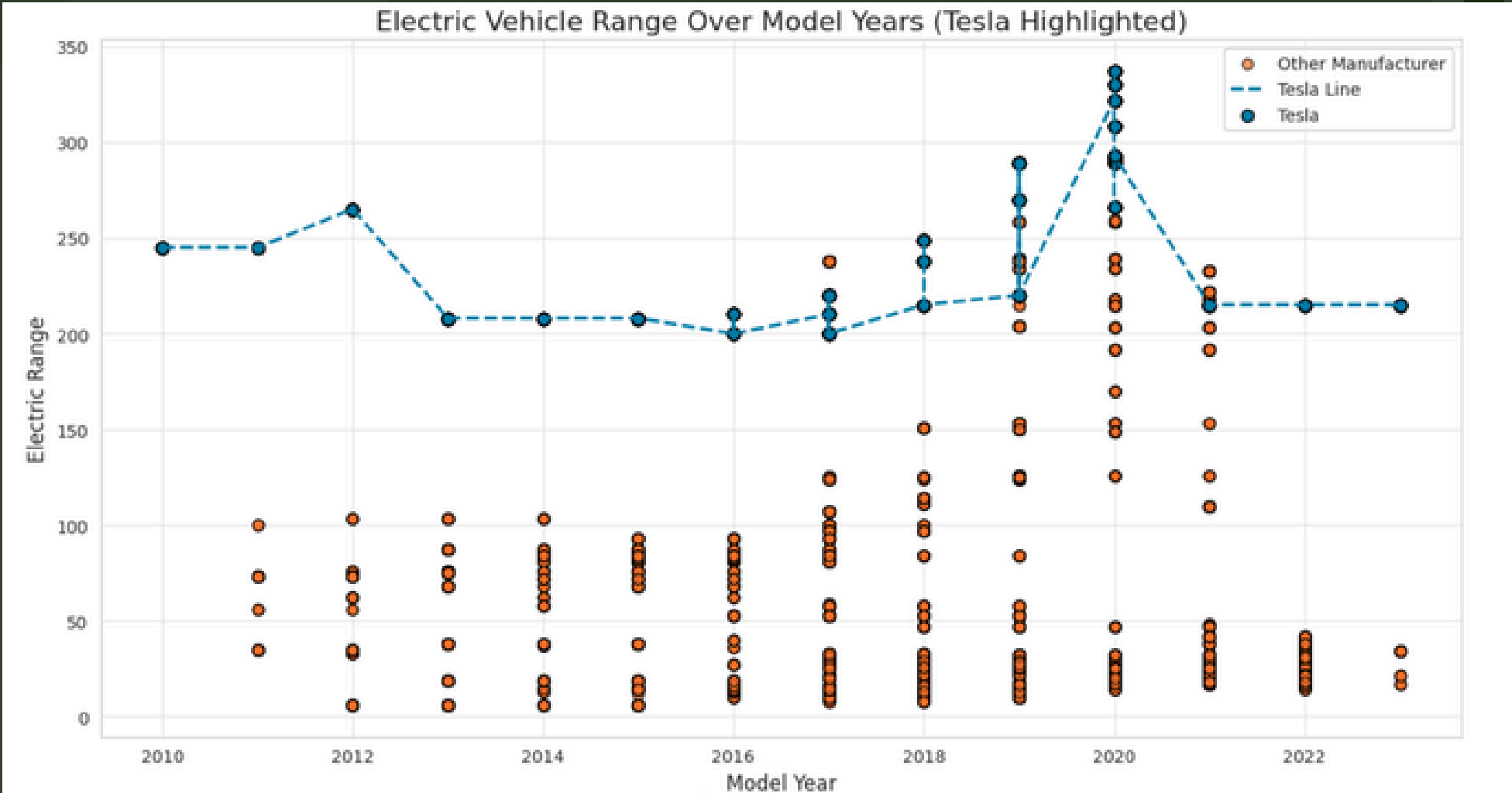
3 ←

```
# Tesla points + line
df_tesla = df_filtered[df_filtered['manufacturer'] == 'Tesla'].sort_values('Model Year')
if not df_tesla.empty:
    plt.plot(
        df_tesla['Model Year'],
        df_tesla['Electric Range'],
        color='#007BA7',
        linestyle='--',
        lw=2,
        label='Tesla Line'
    )
```

Relational Graphs - scatter plots with trend lines

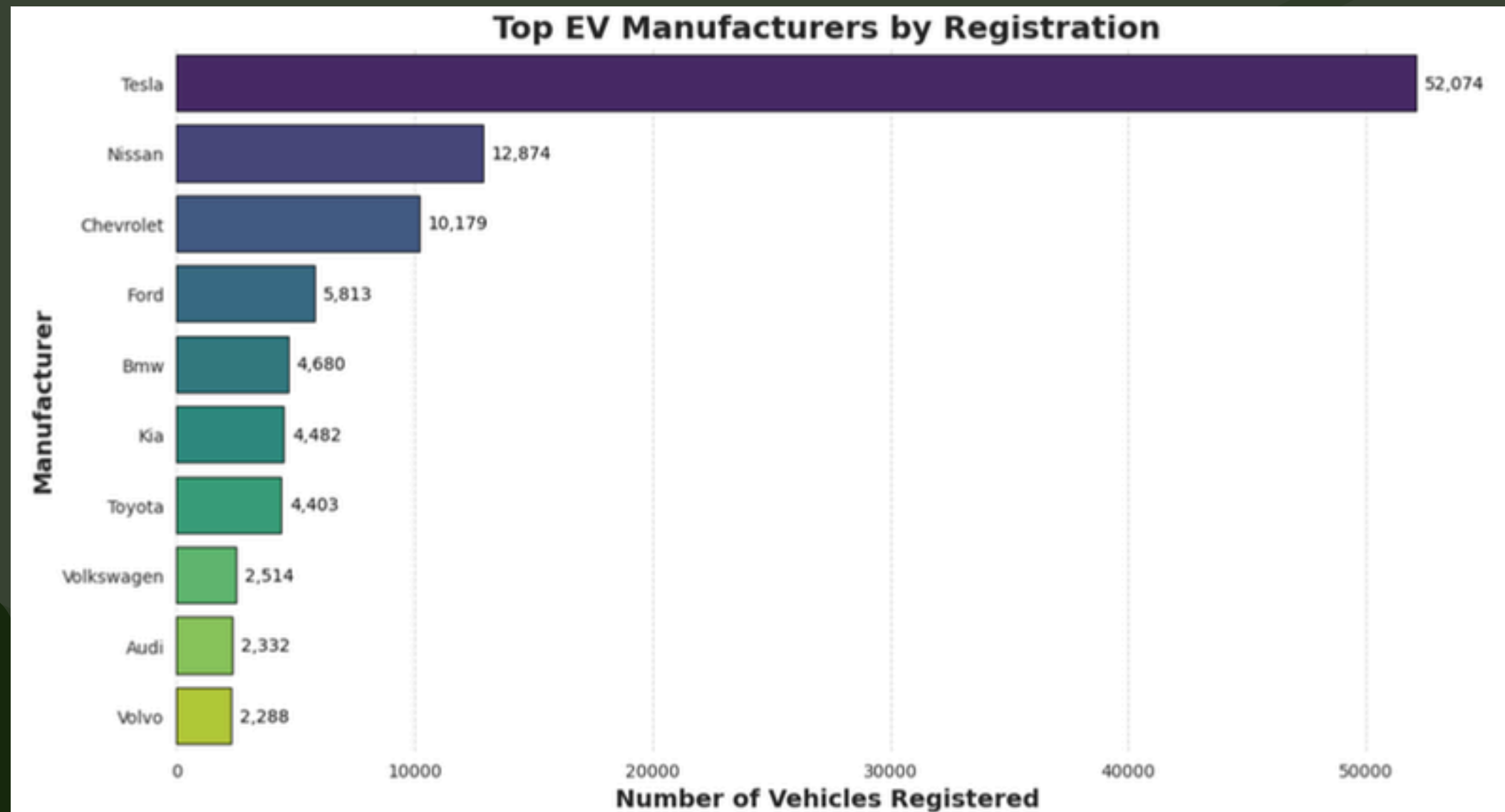
Key Insights :

- **Tesla Leads:** The trend line shows a steady increase in electric range over the years, peaking at 337 miles in 2021, higher than any other manufacturer.
- **Maximum Ranges:**
Tesla → 337 miles (2021)
- **Trend Gap:** Tesla's range steadily increases; other brands show limited improvement.
- **Other Manufacturers:** Most non-Tesla brands have ranges mostly below 150–250 miles, showing slower improvements over the years.
- **Tesla Trend Pattern:** Piecewise-linear with periods of steady range followed by jumps, showing gradual improvements punctuated by major advancements.
- **Market Impact:** The large gap between Tesla and other manufacturers highlights its technological advantage and sets a benchmark for EV range.



Aspect	Relationship	Strength	Type
Tesla	Strong positive trend (range increases with year)	Strong correlation	Piecewise-linear with upward jumps
Other Manufacturers	Mostly flat to mildly positive	Weak to moderate	Linear or flat with minimal growth

Categorical Graphs - Bar plots



```
for patch in ax.patches:
    width = patch.get_width()
    y_pos = patch.get_y() + patch.get_height() / 2
    ax.annotate(
        f'{int(width):,}',
        (width, y_pos),
        ha='left',
        va='center',
        fontsize=10,
        color='black',
        xytext=(5, 0),
        textcoords='offset points'
    )
```

3↑

1↓

```
# Aggregate vehicle counts per manufacturer
top_makes = (
    df_top.groupby('manufacturer')
    .size()
    .reset_index(name='Number of Vehicles')
    .sort_values(by='Number of Vehicles', ascending=False)
)
```

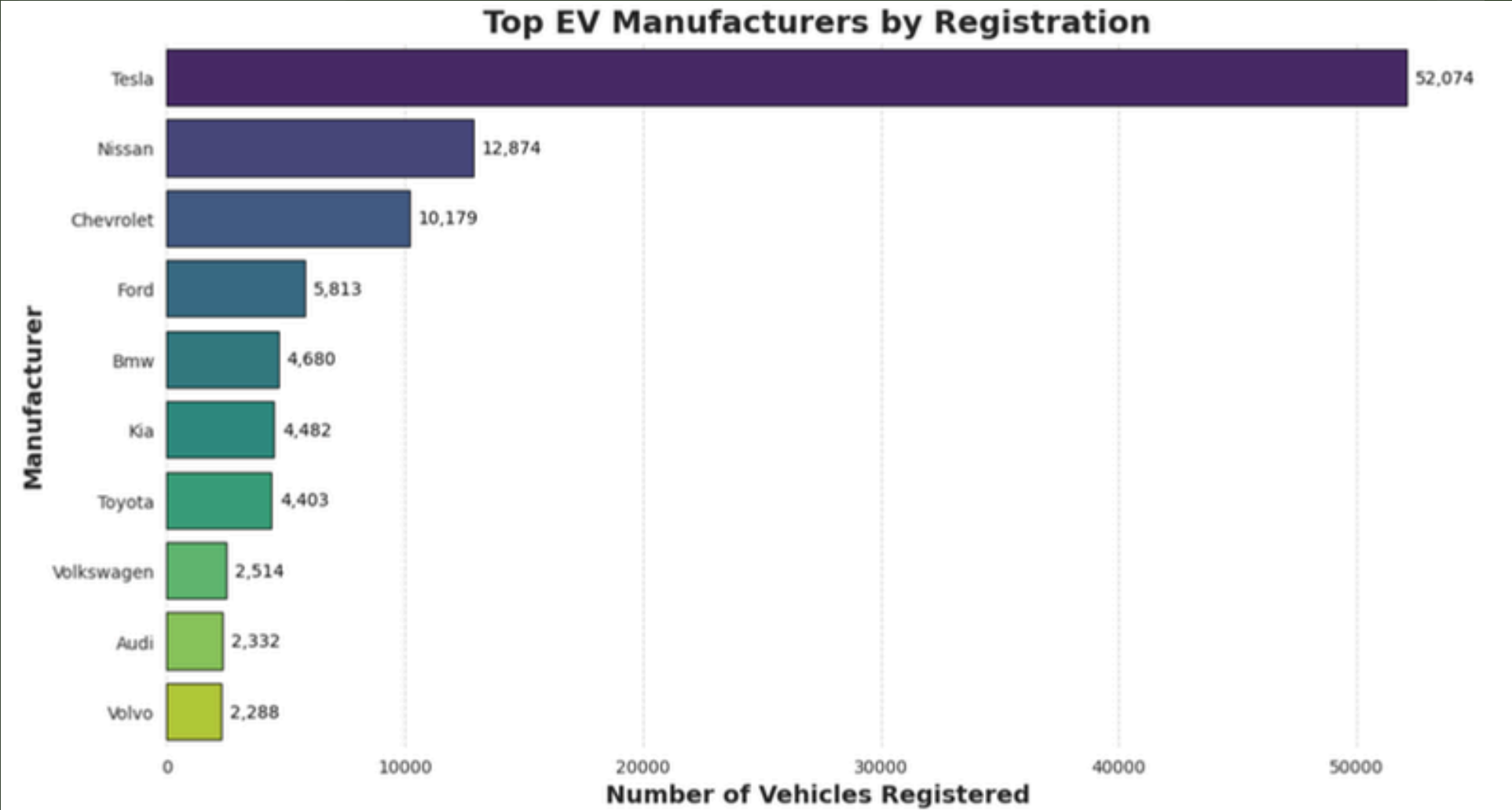
2↓

```
ax = sns.barplot(
    data=top_makes,
    x='Number of Vehicles',
    y='manufacturer',
    order=top_makes['manufacturer'],
    palette='viridis',
    edgecolor='black',
    linewidth=0.7
)
```

Categorical Graphs - Bar plots

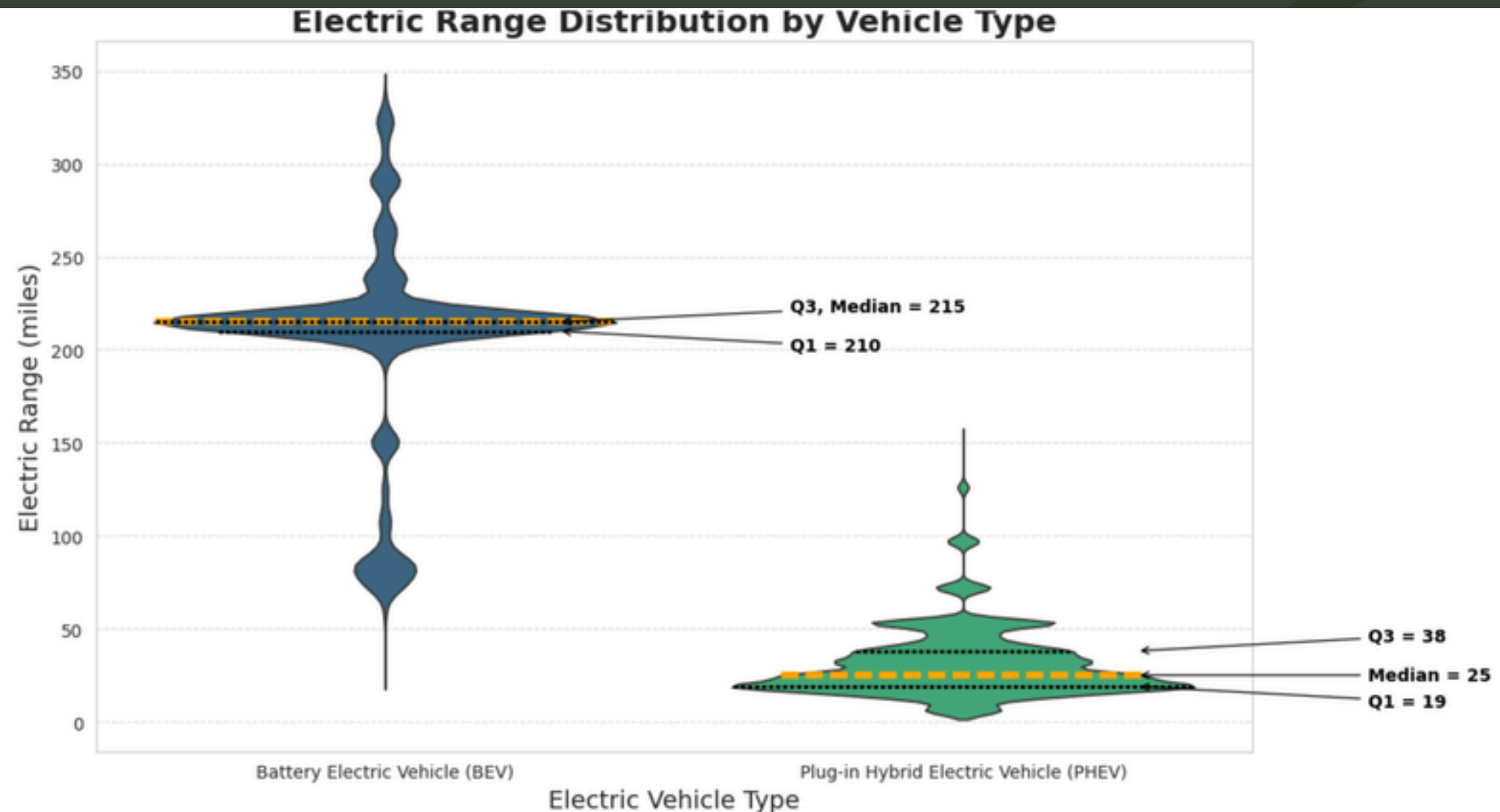
Key Insights

- **Tesla Dominates:** Tesla has the largest number of EV registrations (52,074 units), vastly exceeding any other manufacturer.
- **Other Top Brands:** Nissan (12,874), Chevrolet (10,179), Ford (5,813), BMW (4,680) follow, showing much lower registration volumes.
- **Competitive Gap:** The difference between Tesla and even the next highest brand (Nissan) is over 39,000 units.
- **Market Impact:** This registration data highlights Tesla's dominant position and widespread adoption compared to other EV brands.
- **Trend:** Registrations drop steeply after Tesla, then gradually across other manufacturers, illustrating Tesla's benchmark status.



Aspect	Relationship	Strength	Type
Tesla	Strong positive trend (range increases with year)	Strong correlation	Piecewise-linear with upward jumps
Other Manufacturers	Mostly flat to mildly positive	Weak to moderate	Linear or flat with minimal growth

Statistical Graphs - Violin plot



① →

```
# Create violin plot with quartiles
ax = sns.violinplot(
    data=df,
    x='Electric Vehicle Type',
    y='Electric Range',
    palette='viridis',
    inner='quartile'
)
```

```
for i, vehicle_type in enumerate(stats.index):
    s = stats.loc[vehicle_type]
    Q1, Median, Q3 = int(s['25%']), int(s['50%']), int(s['75%'])

    # Handle overlapping values
    annotations = {}
    if Q1 == Median:
        annotations[f"Q1, Median"] = Q1
        if Q3 != Median:
            annotations["Q3"] = Q3
    elif Q3 == Median:
        annotations["Q1"] = Q1
        annotations[f"Q3, Median"] = Q3
    else:
        annotations = {"Q1": Q1, "Median": Median, "Q3": Q3}

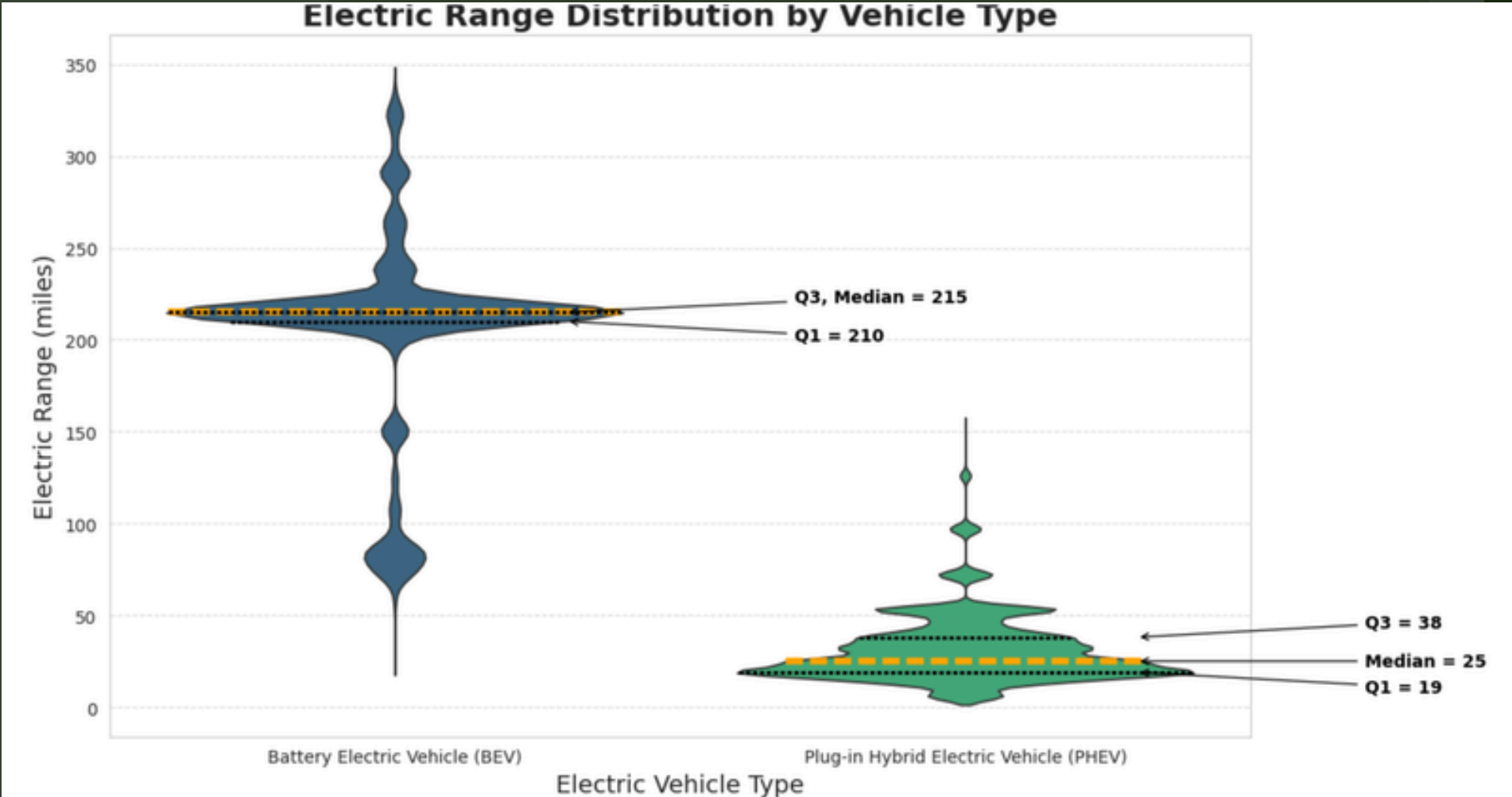
    for name, y in annotations.items():
        offset = y_offsets.get(name.split(',')[0], 0)
        plt.annotate(
            text=f"{name} = {y}",
            xy=(i + 0.30, y),
            xytext=(i + 0.70, y + offset),
            arrowprops=dict(arrowstyle='->', color='black', lw=0.9),
            color='black',
            fontweight='bold',
            va='center'
        )
```

② ↑

Statistical Graphs - Violin plot

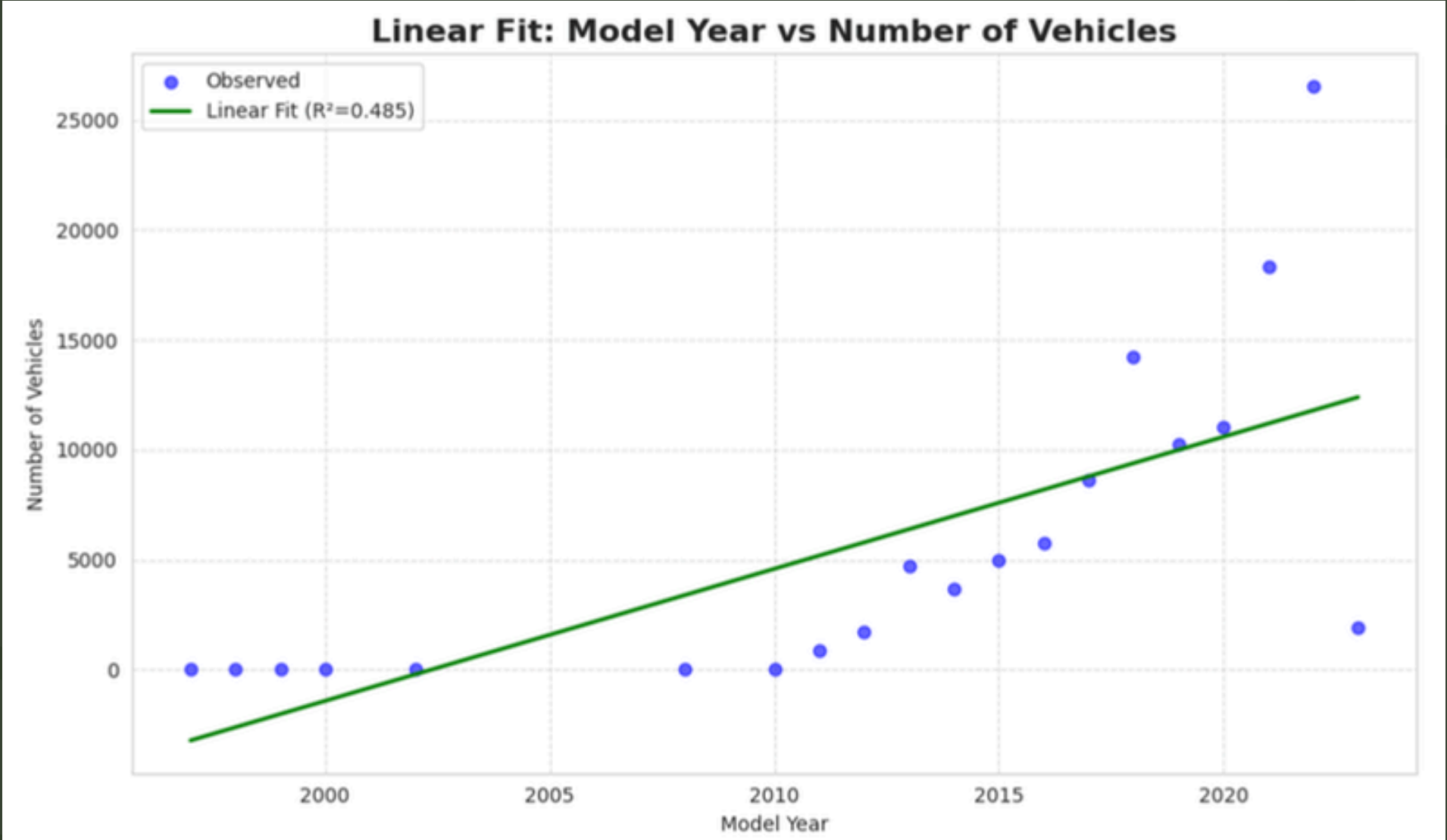
Key Insights

- **BEV Advantage:** BEVs have a much broader and higher range distribution, with modes near 210 miles and 215 miles.
- **PHEV Limitations:** PHEVs mostly cluster between 19 and 38 miles, showing much lower ranges than BEVs.
- **Maximum Ranges:**
BEV: Peaks at 215 miles
PHEV: Peaks at 38 miles
- **Distribution Pattern:** BEV range is spread across multiple levels, while PHEV range is tightly packed at the low end.
- **Market Impact:** BEVs offer far superior range than PHEVs, illustrating BEVs' suitability for longer driving needs and PHEVs for short commutes



Type	Relationship	Strength	Type of Distribution
BEV	Wide, high distribution of ranges	Strong difference between min/max	Multi-modal clusters across range spectrum
PHEV	Narrow, concentrated at lower ranges	Moderate, most ranges below 50 miles	Clustered near low values, right skew

Line fitting - Linear fit



Aspect	Description
Relationship	Positive trend: More vehicles produced in recent years
Strength	Moderate correlation ($R^2 = 0.485$)
Type	Linear fit (steady upward slope)

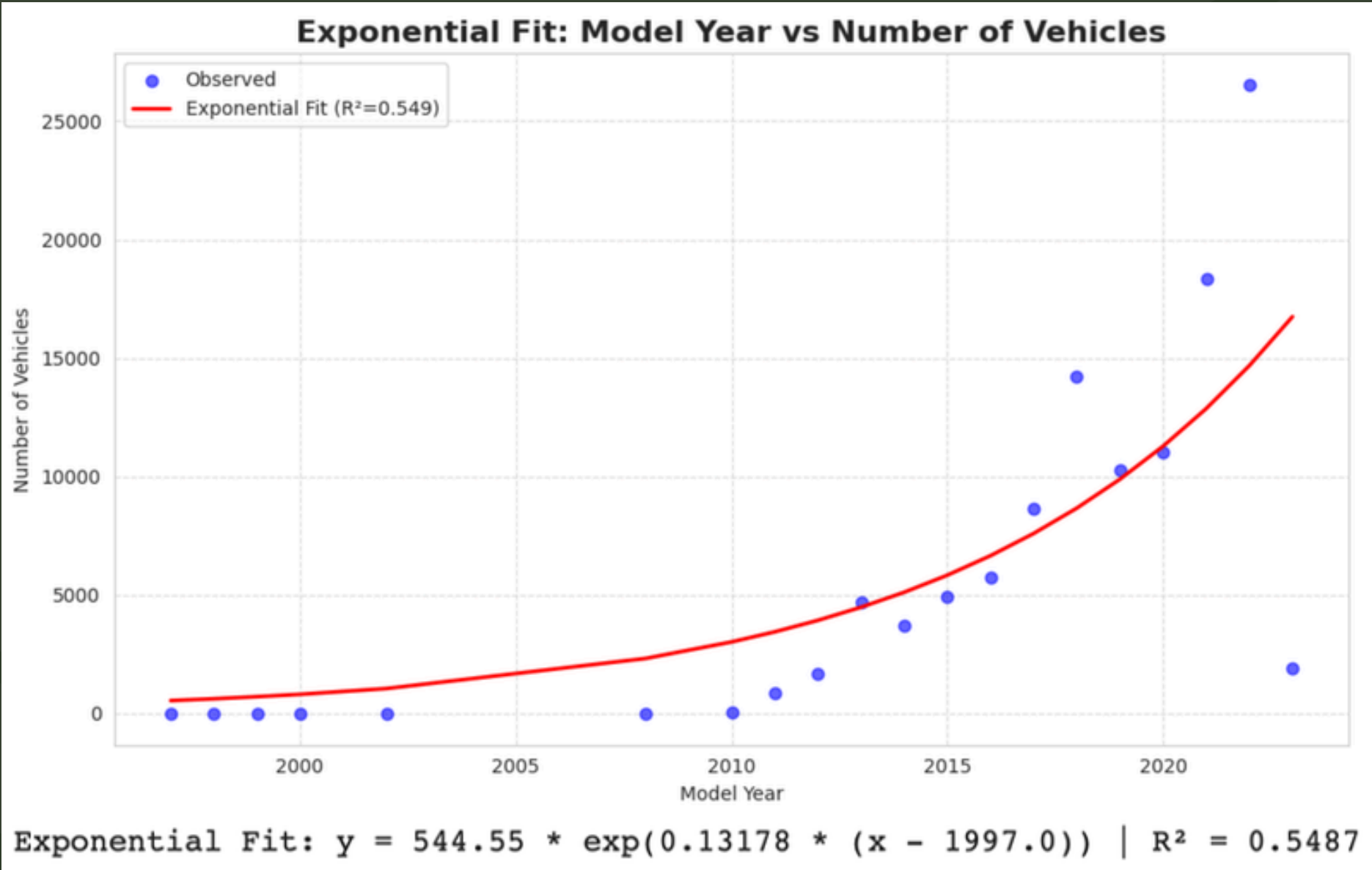
Key Insights

- **Upward Trend:** Vehicle production or registration has generally increased as model years progress.
- **Fit & Strength:** The linear regression line (green) shows a moderate relationship ($R^2 = 0.485$) between year and vehicle count.
- **Equation:** $y = 600.76 \times x - 1,202,947.08$
- **Data Spread:** Recent years, especially after 2015, show larger increases and more variability around the trend.
- **Interpretation:** While growth is apparent, the moderate R^2 indicates other factors besides year also affect vehicle numbers.

Linear Fit: $y = 600.76 * x + -1202947.08$ | $R^2 = 0.4851$

```
# Compute linear regression
slope, intercept, r_value, _, _ = stats.linregress(x, y)
y_linear = slope * x + intercept
r2_linear = r_value ** 2
```


Line fitting - Exponential fit



```
# Define exponential function
def exp_func(x, a, b):
    return a * np.exp(b * x)

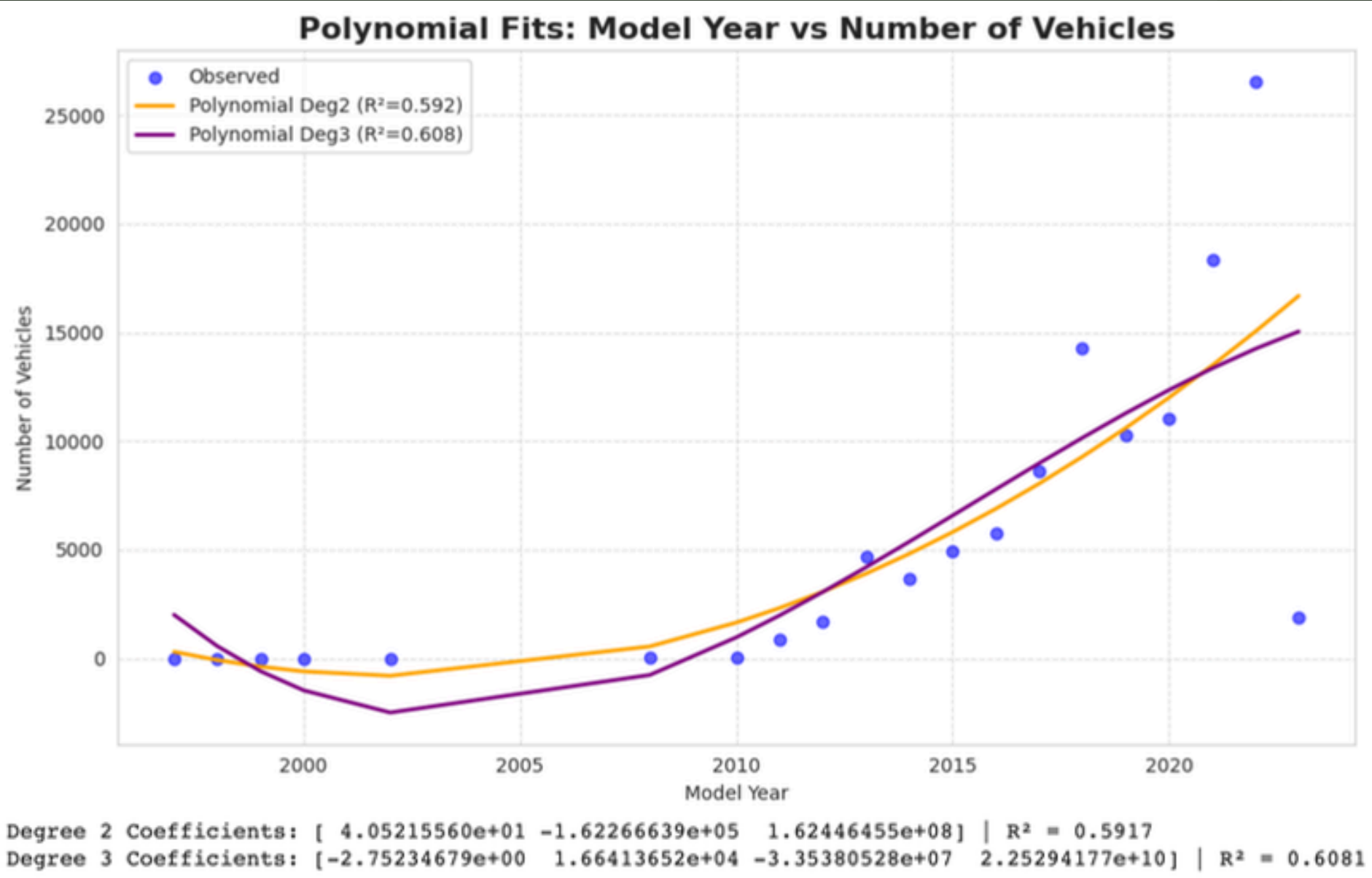
# Fit exponential curve
popt, _ = curve_fit(exp_func, x_norm, y, p0=(y[0], 0.01), maxfev=10000)
y_exp = exp_func(x_norm, *popt)
a_fit, b_fit = popo
r2_exp = r2_score(y, y_exp)
```

Aspect	Description
Relationship	Strong positive exponential trend: vehicle count rises rapidly
Strength	Moderate-to-strong ($R^2 = 0.549$)
Type	Exponential fit (accelerating upward curve)

Key Insights

- **Rapid Growth:** Number of vehicles increases slowly before 2010, then grows sharply in recent years.
- **Fit & Strength:** Exponential regression (red curve) captures the accelerating rise best; correlation is moderate-to-strong ($R^2 = 0.549$).
- **Equation:** $y = 544.55 \times \exp(0.13178 \times (x - 1997.0))$
- **Implication:** EV adoption is accelerating, with growth rate higher than what a linear model predicts.
- **Interpretation:** Recent years show much greater production/registration, suggesting expanding EV market momentum.

Line fitting - Polynomial fit



Aspect	Description
Relationship	Strong positive trend, acceleration is captured by polynomial models
Strength	Polynomial deg 2: $R^2 = 0.592$, deg 3: $R^2 = 0.608$ (best fit in this set)
Type	Polynomial curves (quadratic, cubic)

Key Insights

- Best Fit:** Polynomial degree 3 (purple) slightly outperforms degree 2 (orange) for fitting vehicle growth.
- Growth Pattern:** Both curves reflect slow changes until 2010, then steep rise afterwards; polynomial captures non-linear acceleration.
- R^2 comparison:** Degree 2: $R^2 = 0.592$. Degree 3: $R^2 = 0.608$.
- Interpretation:** Electric vehicle numbers rise slowly early on, then accelerate, with polynomial fits best matching observed data.
- Market Trend:** Higher-order fits show recent advances and suggest continued fast growth in future years.

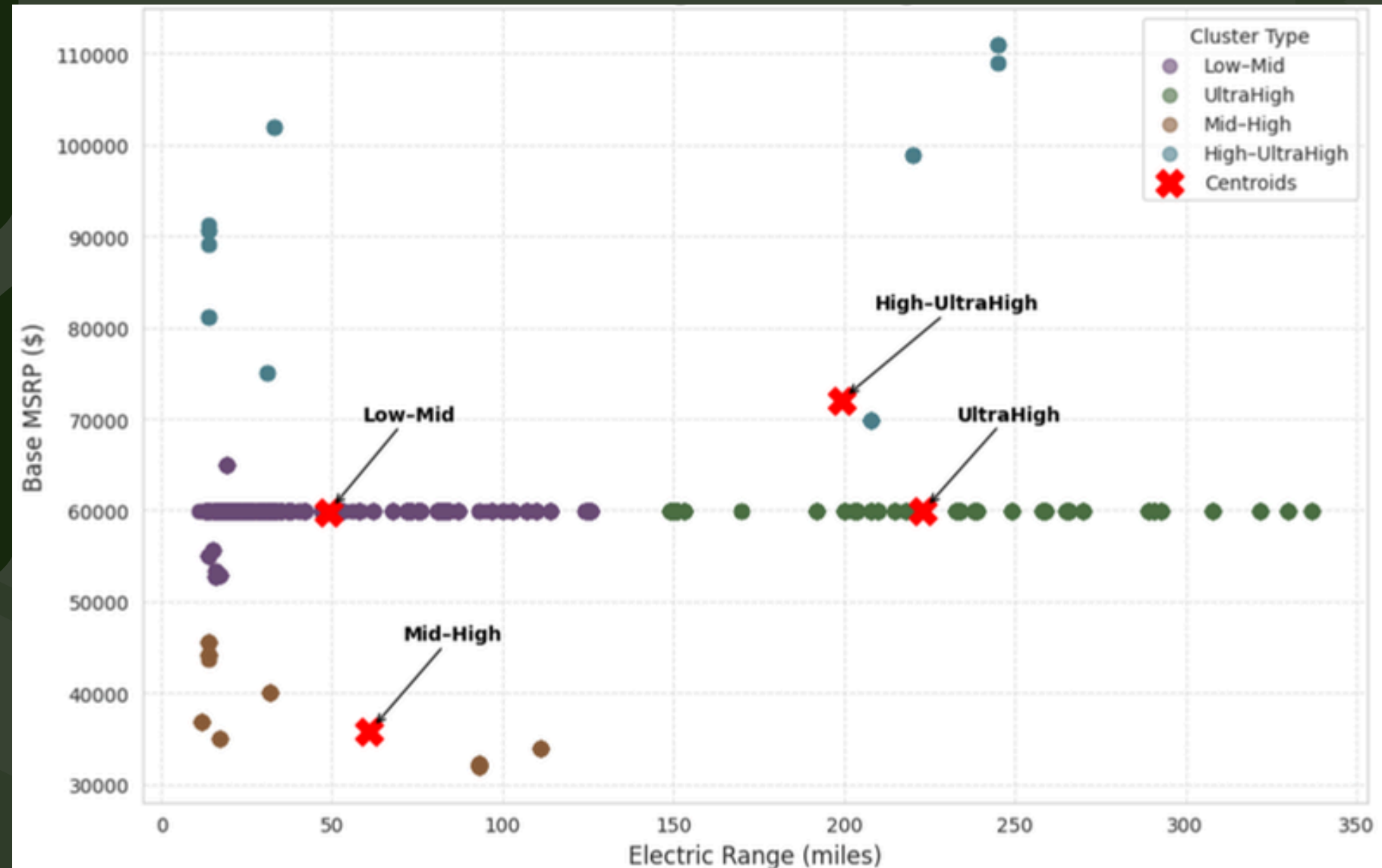
Degree 2 polynomial

```
coeffs_deg2 = np.polyfit(x, y, deg=2)
y_deg2 = np.polyval(coeffs_deg2, x)
r2_deg2 = r2_score(y, y_deg2)
```

Degree 3 polynomial

```
coeffs_deg3 = np.polyfit(x, y, deg=3)
y_deg3 = np.polyval(coeffs_deg3, x)
r2_deg3 = r2_score(y, y_deg3)
```

K-Means Clustering



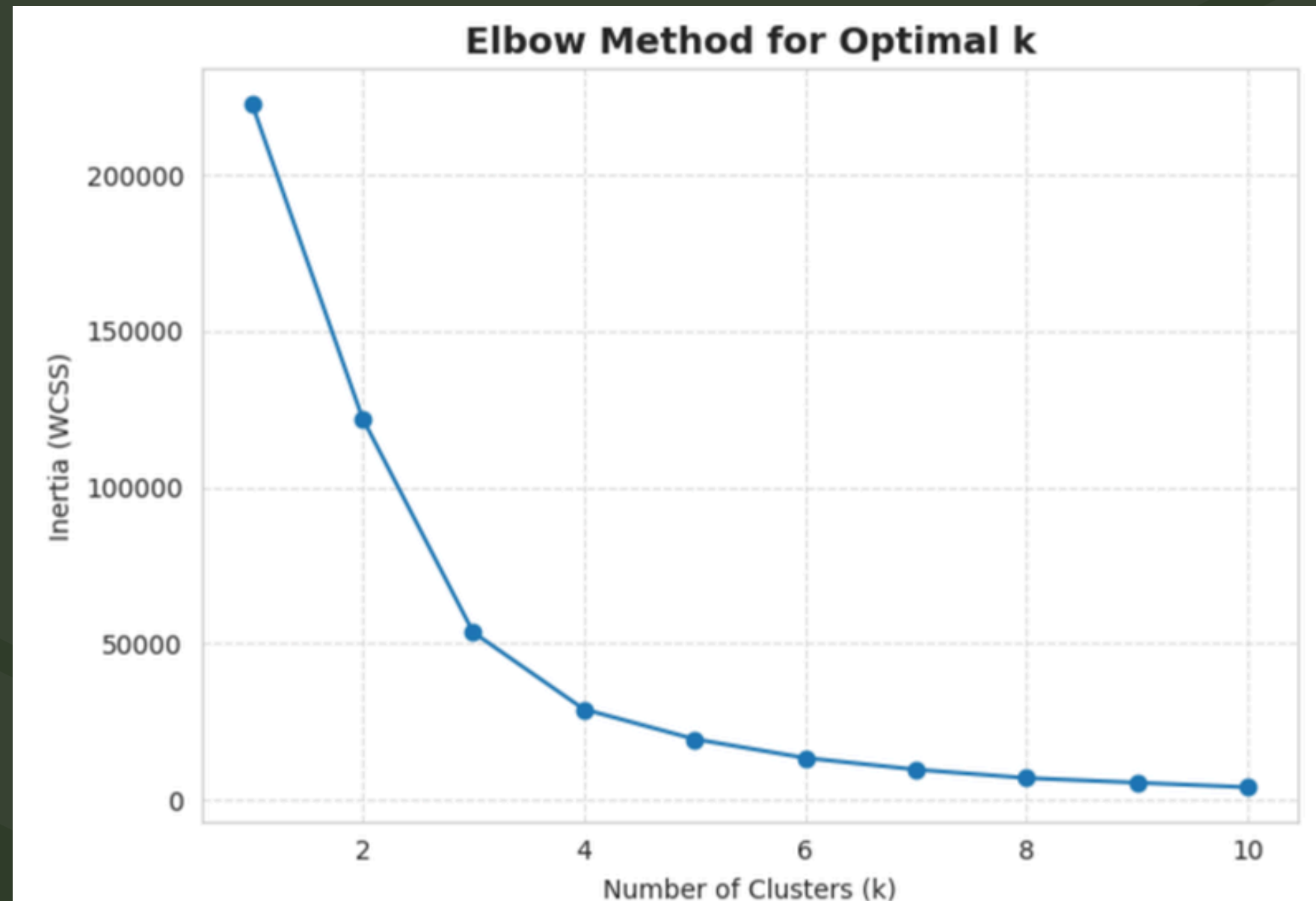
```
# Assign progressive descriptive labels
cluster_labels_desc = {}
for cluster in cluster_summary.index:
    mean_range = cluster_summary.loc[cluster, ('Electric Range', 'mean')]
    if mean_range < 50:
        desc = "Low-Mid"
    elif mean_range < 100:
        desc = "Mid-High"
    elif mean_range < 200:
        desc = "High-UltraHigh"
    else:
        desc = "UltraHigh"
    cluster_labels_desc[cluster] = desc
```

```
# Fill NaN with median
df_clustering['Base MSRP'].fillna(df_clustering['Base MSRP'].median(), inplace=True)
df_clustering['Electric Range'].fillna(
    df_clustering.groupby('Electric Vehicle Type')['Electric Range'].transform('median')
    inplace=True
)
```

Key Insights:

- **Distinct Market Segments:** Four clear clusters—Low-Mid, Mid-High, High-UltraHigh, UltraHigh—based on electric range and base price.
- **Cluster Centroids:** Red X marks indicate each cluster's average range and price for quick visual reference.
- **Market Spread & Guidance:** EVs are unevenly distributed, helping buyers/manufacturers identify suitable segments (budget, mid-market, or luxury).
- **Median Imputation:**
 1. Base MSRP: Overall median fills zeros.
 2. Electric Range: Type-specific median fills gaps
- Ensures clustering quality, prevents bias, and preserves realistic segment structure.

K-Means Clustering



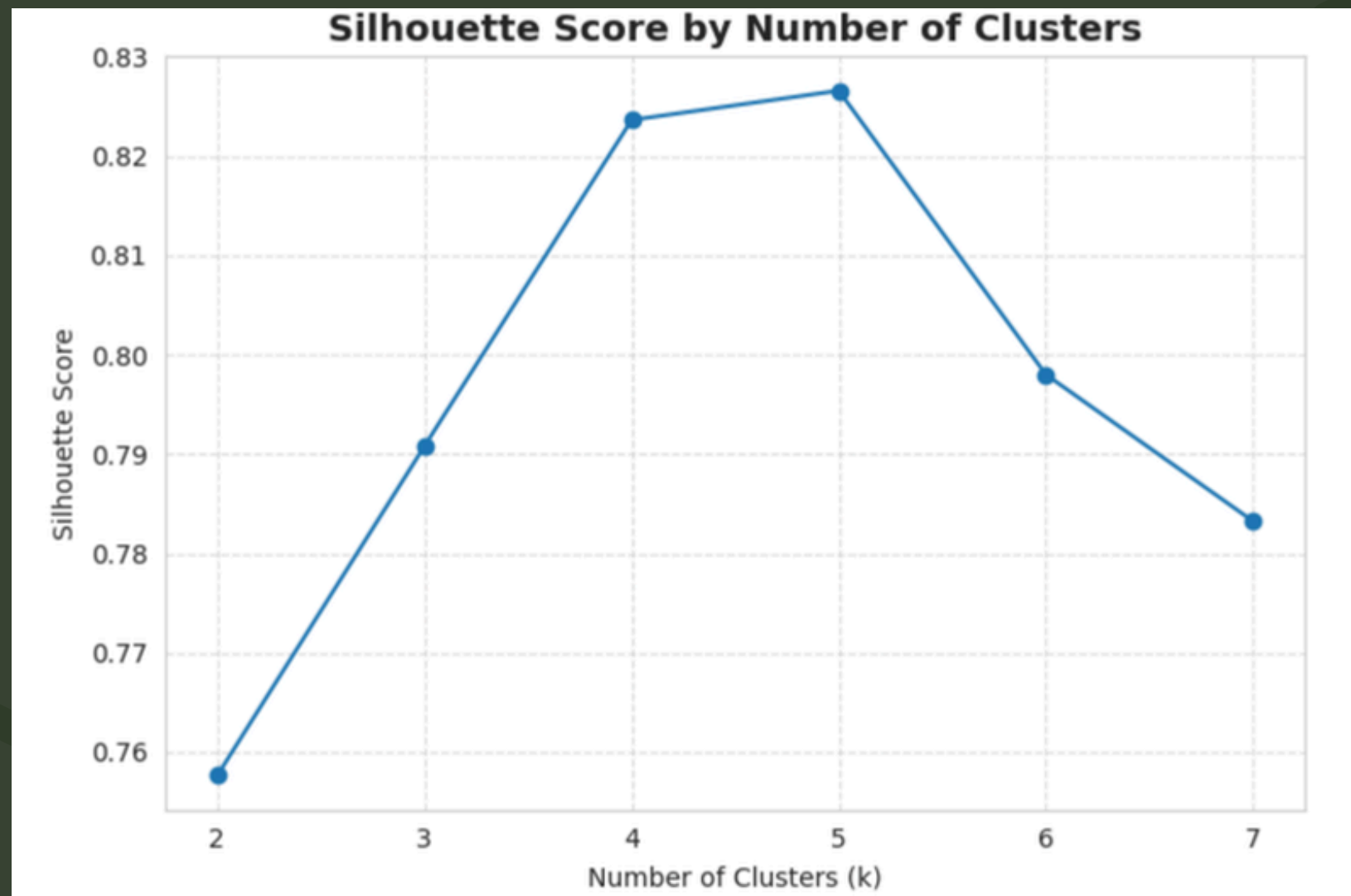
```
inertias, ks = [], list(range(k_min, k_max + 1))

for kval in ks:
    km = KMeans(n_clusters=kval, n_init=10, random_state=42).fit(X_scaled)
    inertias.append(km.inertia_)
```

Key Insights:

- **Sharp WCSS drop (k=1→3):** Inertia falls rapidly from 222,914 to 53,579, showing strong gains in cluster compactness.
- **Elbow point (k=3–4):** Curve flattens after k=4 (inertia $\approx$ 28,953), indicating diminishing returns beyond this point.
- **Marginal gains beyond k=4:** Additional clusters ($k \geq 5$) slightly reduce inertia (e.g., $k=5 \rightarrow 19,386$), risking overfitting or splitting meaningful clusters.
- **Interpretation:** 3–4 clusters capture main variability and provide clear market segmentation.
- **Takeaway:** Limiting to 3–4 clusters keeps the model simple and results interpretable, balancing accuracy and complexity.

K-Means Clustering



Key Insight:

- **Peak at k=5:** Silhouette score rises from 0.758 (k=2) to 0.827 at k=5, indicating best-defined clusters.
- **Optimal clustering:** High scores at k=4 (0.824) and k=5 (0.827) show both options yield well-separated clusters.
- **Diminishing returns beyond k=5:** Scores drop at k=6 (0.798) and k=7 (0.783), suggesting over-splitting and lower cohesion.
- **Interpretation:** Combined with elbow method, k=4 or k=5 captures balanced, interpretable clusters.
- **Takeaway:** Adding more clusters beyond this reduces cluster quality and clarity.

```
results = []
for kval in k_values:
    if kval < 2:
        results.append((kval, np.nan))
        continue
    km = KMeans(n_clusters=kval, n_init=10, random_state=42)
    labels = km.fit_predict(X_scaled)
    sil = silhouette_score(X_scaled, labels)
    results.append((kval, sil))
return results
```


Conclusion



Market Overview

Tesla's Lead:
Tesla offers the highest electric range and dominates adoption and technology.

Market Growth:
EV registrations have accelerated rapidly, especially after 2015, showing strong market expansion.

Trend Analysis

Accelerating Growth:
EV registrations have risen rapidly, with exponential/polynomial fits showing steep increases after 2015.

BEVs Outperform:
Battery EVs provide higher ranges, making them more suitable for long-distance travel.

Clustering Insights

Optimal Clusters:
Both elbow and silhouette methods indicate 3–4 clusters are ideal for segmentation.

Practical Use:
Clustering highlights meaningful groupings without creating overly granular or small clusters.

Thank You!

